Grid nesting

palmgroup

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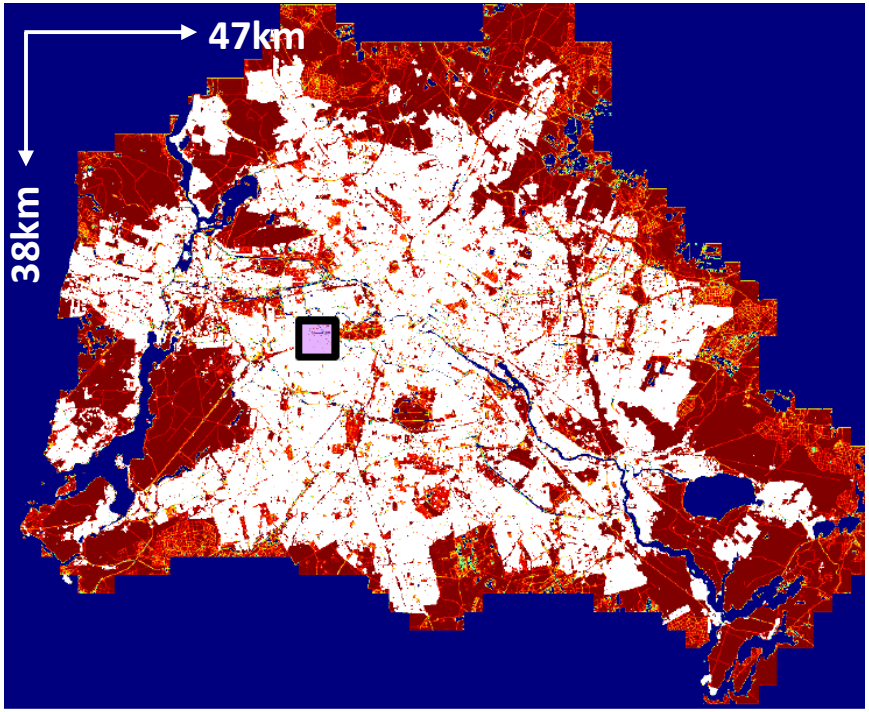
Why is there a need for a grid nesting?

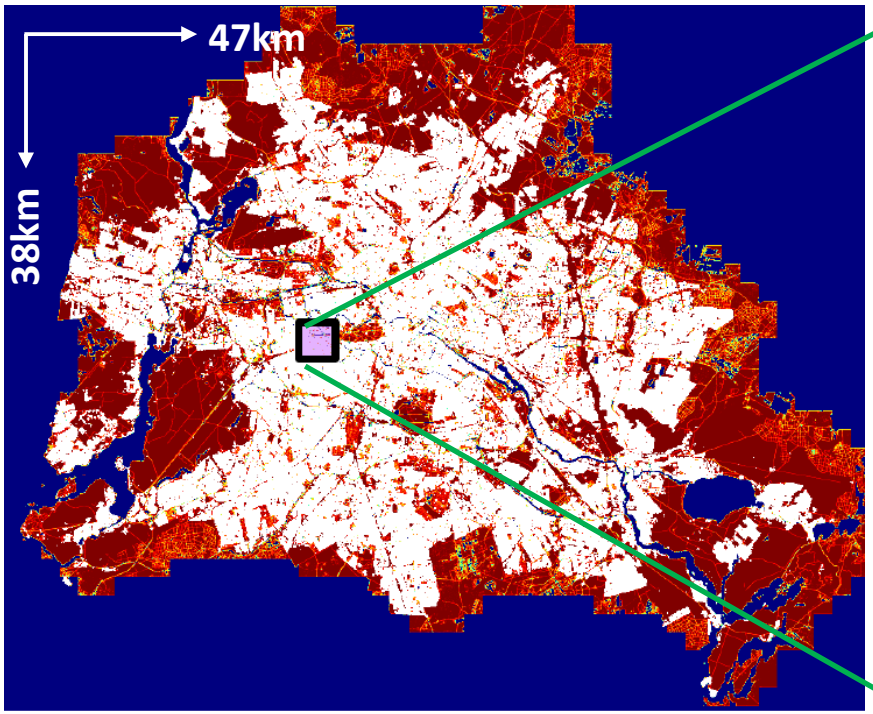


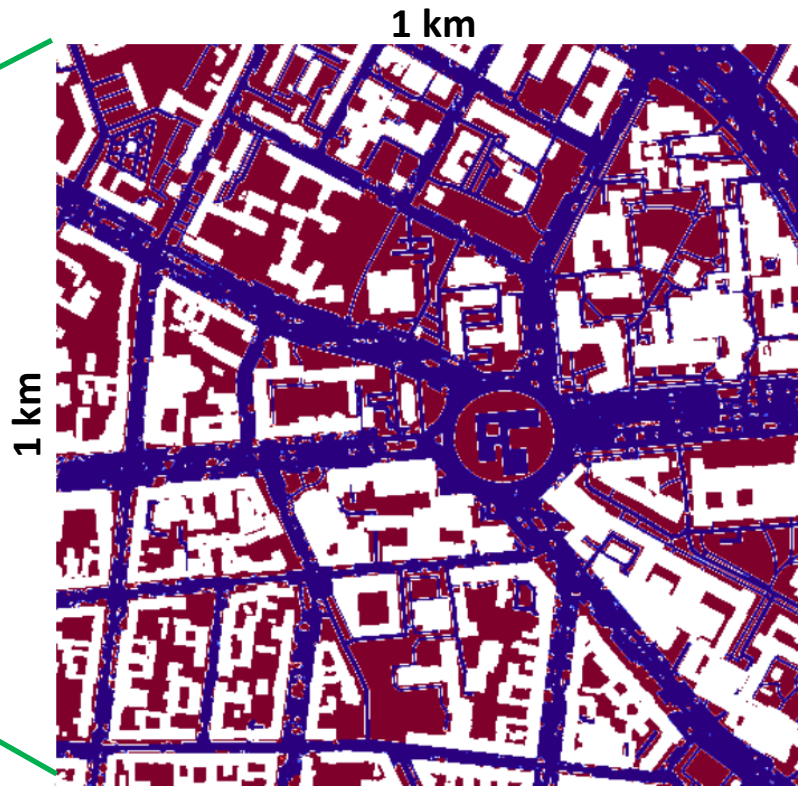
City of Berlin

- **10 m resolution: 10^9-10^{10} grid points**
- **1 m resolution: 10^{12}-10^{13} grid points**

Not feasible for detailed parameter studies, even with 10 m resolution everywhere!

Why is there a need for a grid nesting?





City of Berlin

- **50 m resolution: 10^7-10^8 grid points**

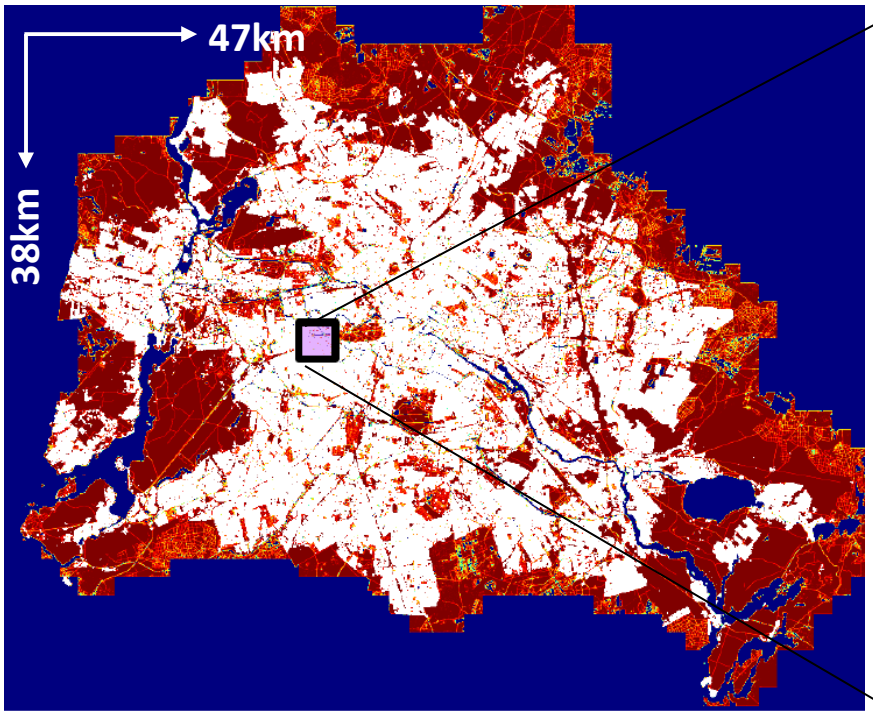
50 m resolution is feasible, but it might be too coarse for using it everywhere!

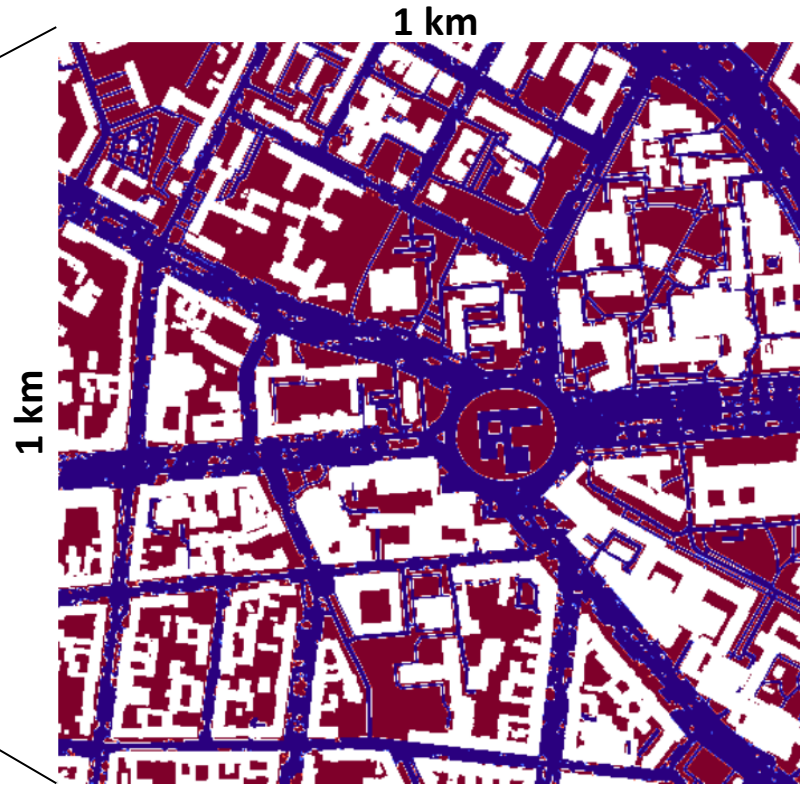
Increase grid resolution for domain of interest

- **1 m resolution: 10^8 grid points**

Attention: Too high grid spacing ratios can create unrealistic results

Why is there a need for grid nesting?





City of Berlin

- **50 m resolution: 10^7-10^8 grid points**

50 m resolution is feasible, but it might be too coarse!

Increase grid resolution for domain of interest

- **1 m resolution: 10^8 grid points**

We call this "self-nesting".

Grid nesting

Motivation

- Boundary layer processes contain a wide range of scales, ranging from the mesoscale, e.g. urban heat island, down to the microscale, e.g. effects of single trees, building and roof shapes, local emissions, etc.

- To consider both large model domains at a small grid size would be required – not feasible today, even not on supercomputers !

- Idea: Consider mesoscale processes on a coarser grid and refine the grid within the domain of interest.

- This is what we call **“mesoscale-nesting”**.

Presentation outline

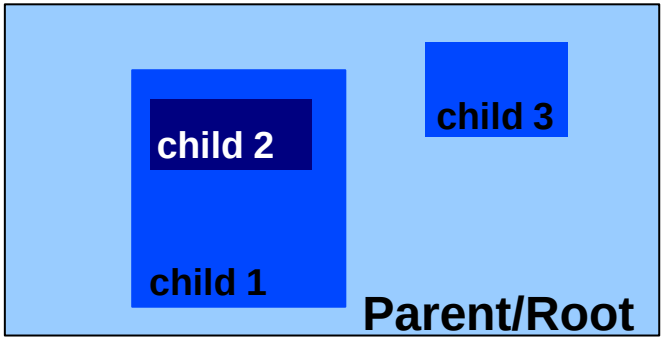
- General/Technical information about the **self-nesting** method
  → PALM's first nesting system
  - Steering
  - Examples
- **Mesoscale (Offline) nesting** in large-scale models (e.g., ICON, WRF)
  → PALM's second nesting system
- Outlook and open points

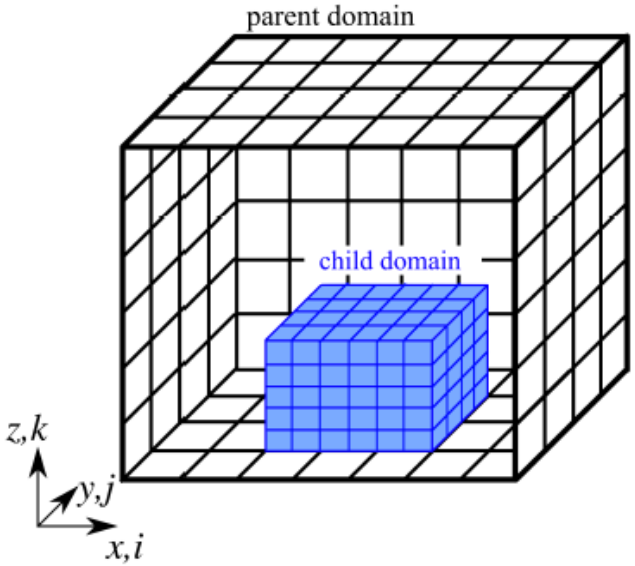
Basics of the nesting

- **Goal:** Reduce computational costs significantly
  - Enable simulations with a large domain and detailed analysis within domain of interest
  - Enable industrial application of LES with PALM (urban environments, site assessment in wind energy)
- **Idea:**
  - High grid resolution within domain of interest
  - Coarse grid resolution of other/outer parts of model domain
- **Precondition/Requirement:**
  - Nested domain results should correspond to results of the “classical” non-nested simulation, where a uniform grid spacing identical to the grid spacing in the nested domain is used in the whole domain

General principles 3D nesting – Domain Structure

- Self-nesting is a PALM-PALM-coupling with two or more simulations running in parallel to each other with a continuous communication at runtime.
- One root domain (outermost and coarsest-resolution LES domain) and up to 63 child domains embedded into the root model are possible.
- Childs can be recursively nested within each other → model domain can be parent and child at the same time (see child 1).
- Child domains can also be parallel to each other sharing same parent domain.





General principles 3D nesting – Domain Structure

- All child domains must be completely inside their parent domain → no overlapping of parallel child domains → they have one parent domain.

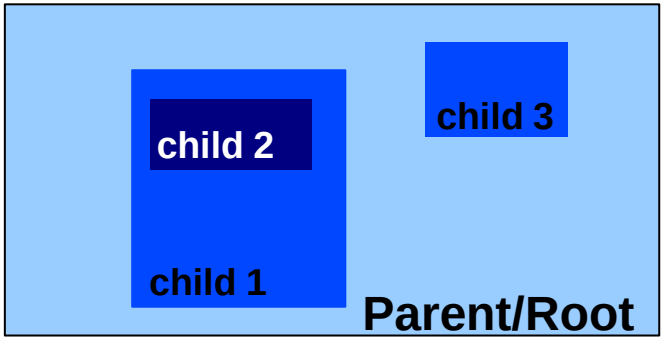
- Lower boundaries of childs can be elevated with respect to their parent.

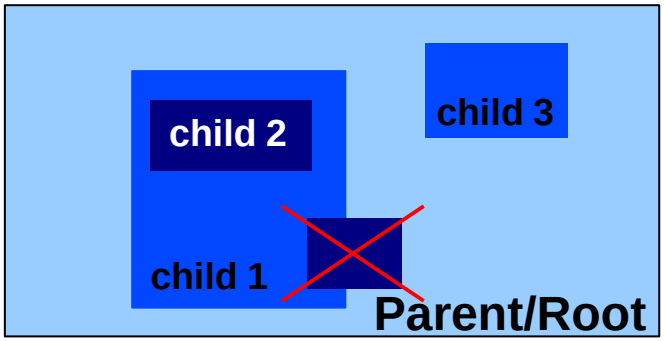
- Outer boundaries of child domain must match the underlying parent grid lines in all directions.

- Inside child domain all parent-grid lines must match the corresponding child-grid lines

→ Grid-spacing ratios in each direction must be integer valued.

→ Vertical grid stretching is only allowed in the root domain above the top level of the highest nested domain

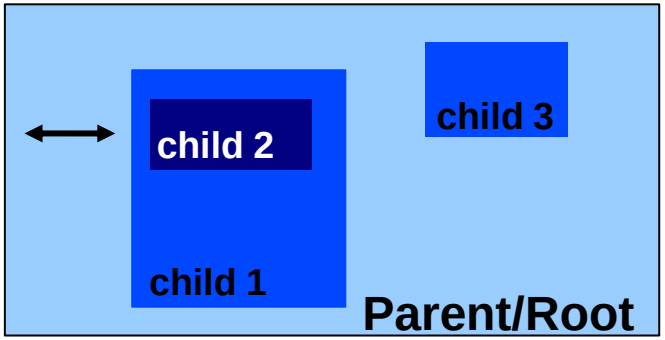


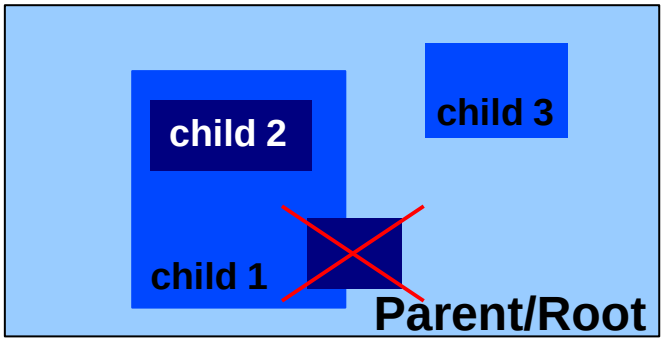


General principles 3D nesting – Domain Structure

- 2D Domain decomposition of child domains during parallelization must be realized in a way that the sub-domain size is never smaller than the parent grid spacing in the respective direction.

- 4 parent grid cells between the boundaries of child and parent domains are necessary.

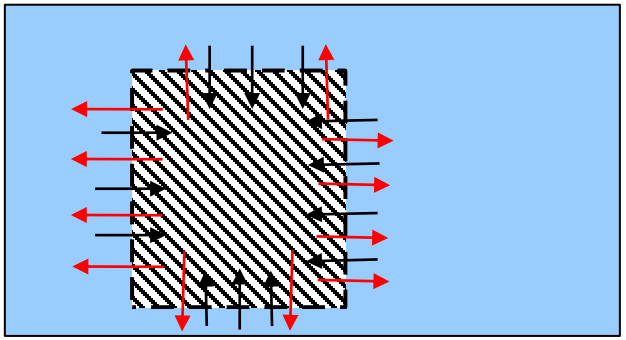




General principles 3D nesting – Data transfer

- Two-way (default mode) or one-way coupling is possible.
- All prognostic variables are coupled except the SGS-TKE *e* (has no real benefit and a coupling of *e* is everything else than straightforward since it strongly depends on grid resolution).
- Most important requirements for the nesting algorithm: Minimizing flux conservation errors and enabling complex topography.

Data transfer between parent child



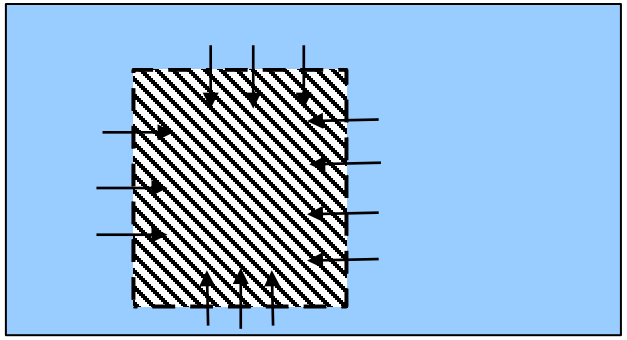
Data in (parent to child)

Data out ? (child to parent)

General principles 3D nesting – Data transfer

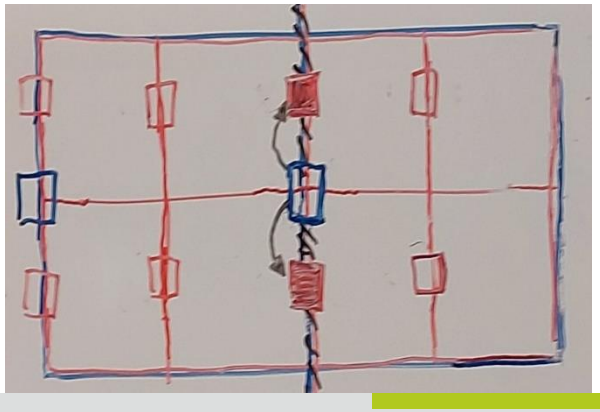
- **Two-way coupling/nesting**:
  - The focus is on both parent and child domain (e.g., dispersion scenarios).
  - Child domain obtains boundary-conditions from parent through zero order or linear interpolation.
  - For boundary-normal velocity components, the original parent values are used (e.g., left child boundary u-component, values are set directly on the boundary → zero order interpolation):

Data transfer



Data in (parent to child)
Interpolation

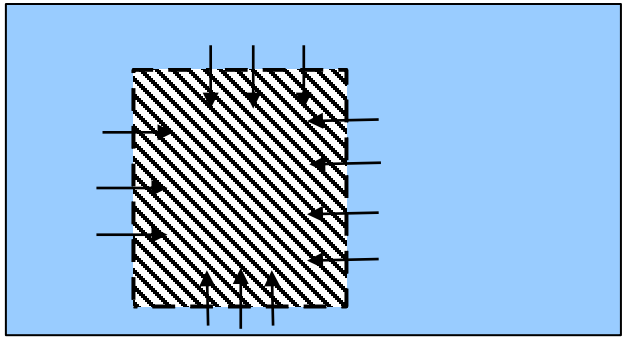
Data out (child to parent)
Anterpolation



General principles 3D nesting – Data transfer

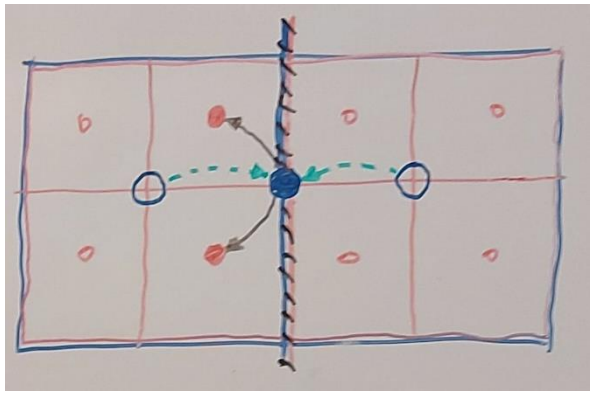
- **Two-way coupling/nesting**:
  - The focus is on both parent and child domain (e.g., dispersion scenarios)
  - Child domain obtains boundary-conditions from parent through zero order or linear interpolation
  - For scalars, averaged parent values from the nest boundary are used (e.g., left child boundary, values are set for the first ghost point based on two surrounding parent grid values → linear interpolation):

Data transfer



Data in (parent to child)
Interpolation

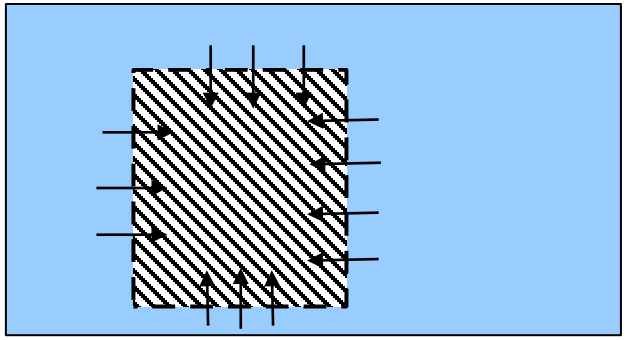
Data out (child to parent)
Anterpolation



General principles 3D nesting – Data transfer

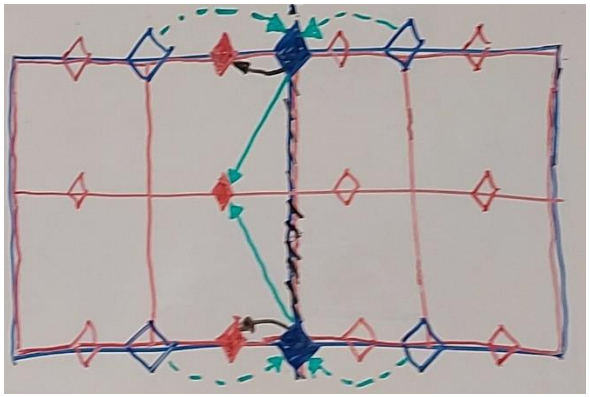
- **Two-way coupling/nesting**:
  - The focus is on both parent and child domain (e.g., dispersion scenarios)
  - Child domain obtains boundary-conditions from parent through zero order or linear interpolation
  - For staggered velocity components with respect to the boundary-normal velocity the following formula is used (e.g., left child boundary v-component, values are set for the first ghost point based on four surrounding parent grid values → linearly interpolated twice or once):

Data transfer



→ Data in (parent to child)
Interpolation

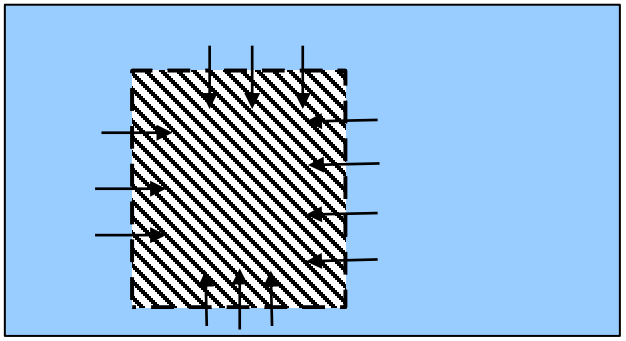
Data out (child to parent)
Anterpolation



General principles 3D nesting – Data transfer

- **Two-way coupling/nesting**:
  - For *e*, a simple Neumann condition (zero-gradient) is used on child domain boundaries.
  - The reason behind the (randomly appearing) interpolation scheme is explained in detail in “A Nested Multi-Scale System Implemented in the Large-Eddy Simulation Model PALM” by Hellsten et al. (2021) and goes far beyond an introduction.

Data transfer



Data in (parent to child)
Interpolation

Data out (child to parent)
Anterpolation

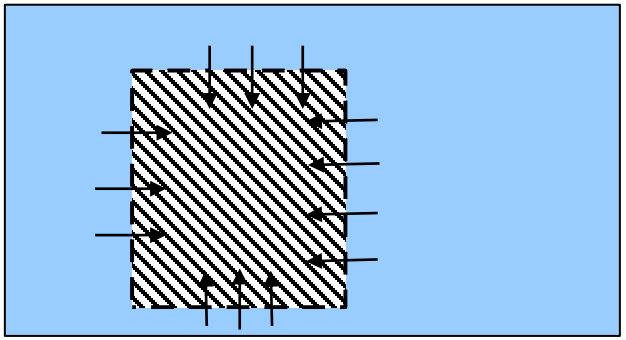
General principles 3D nesting – Data transfer

- **Two-way coupling/nesting**:
  - Child influences parent through anterpolation.
  - Mapping the fine-grid solution back to the parent domain.
  - Averaging over one parent-domain grid volume around the parent grid node of the variable in question (i.e., top-hat filtering)

$$\widehat{\phi}_{I,J,K} = \frac{1}{N_{I,J,K}} \sum_{i_1(I)}^{i_2(I)} \sum_{j_1(J)}^{j_2(J)} \sum_{k_1(K)}^{k_2(K)} \phi_{i,j,k}$$

  - Buffer zones of two prognostic grid points, where the anterpolation is omitted, are applied next to the child-domain boundaries (except bottom boundary) to avoid an unstable feedback loop between anterpolation and interpolation.

Data transfer



→ Data in (parent to child)
Interpolation

Data out (child to parent)
Anterpolation

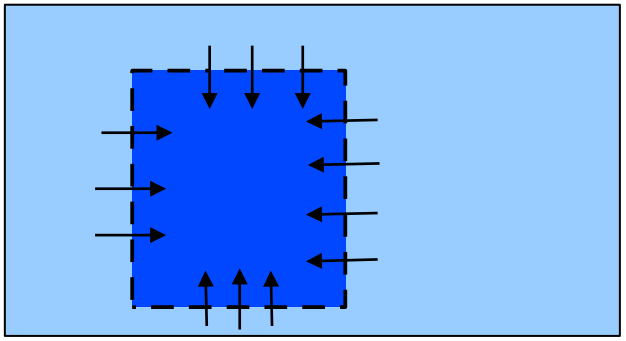
→ The results must always be examined regarding unphysical feedbacks between parent and child!

General principles 3D nesting – Data transfer

- **One-way coupling/nesting**:
  - The focus is only on the child domain (e.g., complex terrain)
  - Anterpolation step is omitted (saves computational costs).
  - Parent simulation is independent from child simulation – no feedback (except time step).
  - Decoupling of turbulence may lead to strong discontinuities.

  → The results of parent and child may become very different from each other but unphysical feedbacks between parent and child are uncritical.

Data transfer

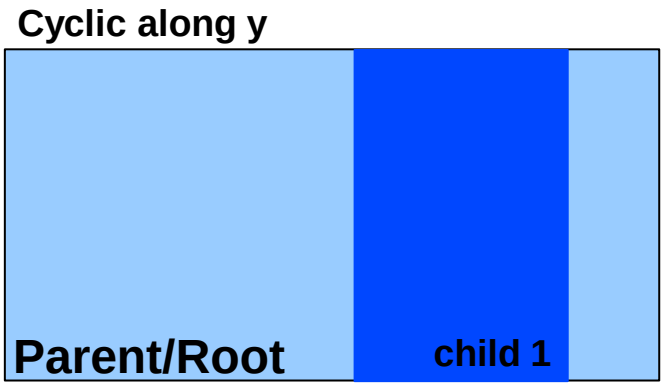


→ Data in (parent to child)
Interpolation

- Coupling data transfer by default at each Runge-Kutta time sub-step just before the pressure solver, independent from the coupling method. Minimum time step of parent/child is used.
- Time step synchronization can be switched-off.

Special setups – Domain Structure

- 2D or even 1D nesting can also be applied.
- 2D: Root domain size exactly matches the extension of the child in one horizontal direction.
- 1D: All domains have the same horizontal extent → pure vertical nesting.
- For pure vertical nesting child can be elevated, e.g. to better resolve the entrainment layer at the top of a convective mixed layer.



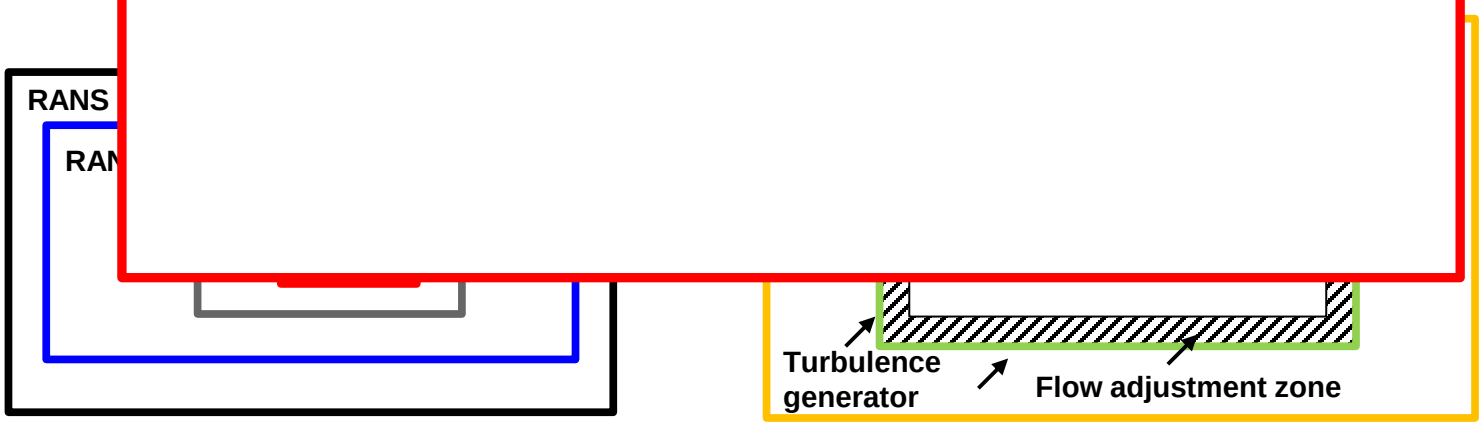
General principles – Initialization, boundary conditions, pressure solver

- Child domain is by default initialized with 3D volume data from parent (`initializing_actions = 'interpolate_from_parent'`), any other initialization, e.g. `'set_constant_profiles'` will be overwritten.
- Boundary conditions at lateral and bottom/top boundaries of nested domains are internally set to `'nested'`.
  - Zero-gradient conditions for pressure (Neumann condition).
  - Dirichlet conditions for prognostic quantities derived from interpolation.
  - For the root domain of a nested run the default is as usual (e.g., `'cyclic'` for lateral boundaries).
  - Exception: pure vertical nesting (lateral boundaries of parent and child are the same), where still cyclic lateral boundary conditions are applied for the child as default.

Nesting for RANS/LES mode

- PALM ca sures (two for
- Nesting
  - RAN
  - LES
  - RAN e turb

RANS/LES nesting is still an experimental feature!



Parameter file

- Each model/domain has its own parameter file:
  → **`<ri>_p3d`** (root domain), **`<ri>_p3d_N02`** (first child), …
- Additional *NAMELIST* group name: **`&nesting_parameters`**
  → provides information about all domains
  → only in **`<ri>_p3d`** of the root domain!
- Other input files (e.g topography, static and dynamic driver) have to be provided for each domain
  → using the domain tags e.g., **`<ri>_static_N02`**, **`<ri>_static_N03`**, …
- Data output for each domain in separate files
  → using the domain tags e.g. **`<ri>_rc`** , **`<ri>_rc_N02`**, ...

Parameter file – example parent (example_p3d)

name id parent_id npe_total lower_left_x lower_left_y vertical shift

```
&nesting_parameters
        domain_layouts = 'palm_coarse', 1, -1, 64, 0.0,   0.0,   0.0,
                         'palm_fine',   2,  1, 64, 320.0, 160.0, 0.0,

        nesting_mode         = 'two-way',
        nesting_bounds       = '3d_nested',
        synchronize_timestep = .FALSE.,

/

&initialization_parameters
    nx = 127, ny = 63, nz = 32,
...
```

Parameter file – example child (example_p3d_N02)

No NAMELIST group &nesting_parameters

```
&nesting_parameters
        domain_layouts = 'palm_coarse', 1,  1, 64, 0.0,   0.0,   0.0,
                         'palm_fine',   2,  1, 64, 320.0, 160.0, 0.0,

        nesting_mode          = 'two-way',
        nesting_bounds        = '3d_nested',
        synchronize_timestep = .FALSE.,

/

&initialization_parameters
    nx = 127, ny = 63, nz = 32,
...
```

Miscellaneous

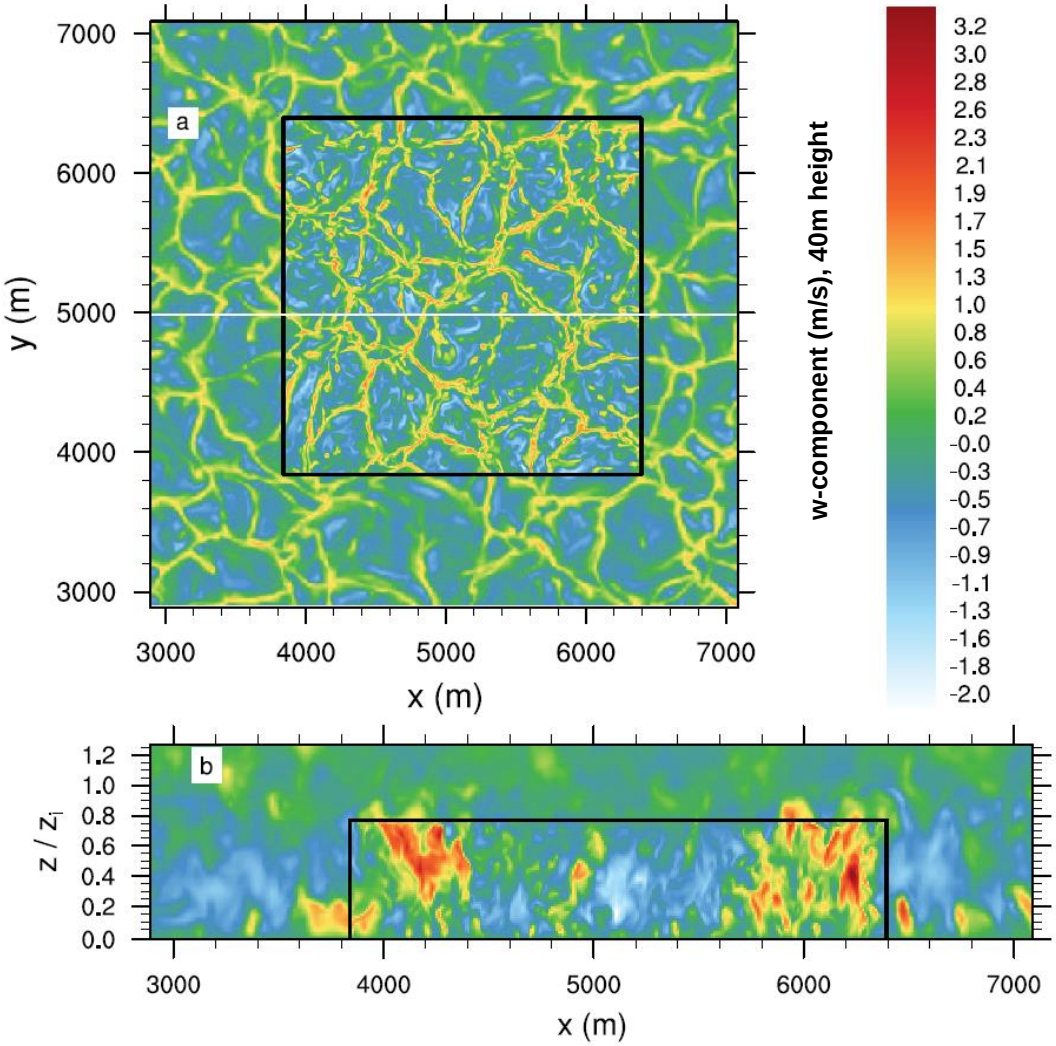
- Assure that the total number of given cores match the sum of cores given for each domain´.
- The parent usually follows the time step of the child. Can be switched-off via **`synchronize_timesteps = .FALSE.`**.
- Take care of a reasonable load balance between child and parent, i.e., subdomain sizes in terms of grid points should be comparable. Otherwise the parent always waits for the child or vice versa. Also non-synchronized timesteps need to be considered.

```
palmrun -r example -a "d3#" -X 128 ...

&nesting_parameters
        domain_layouts = 'palm_coarse', 1, -1, 64, 0.0,   0.0,    0.0
                         'palm_fine',   2,  1, 64, 320.0, 160.0,  0.0,

        nesting_mode           = 'two-way',
        nesting_bounds         = '3d-nested',
        synchronize_timesteps  = .FALSE.,
/
```

Three Examples

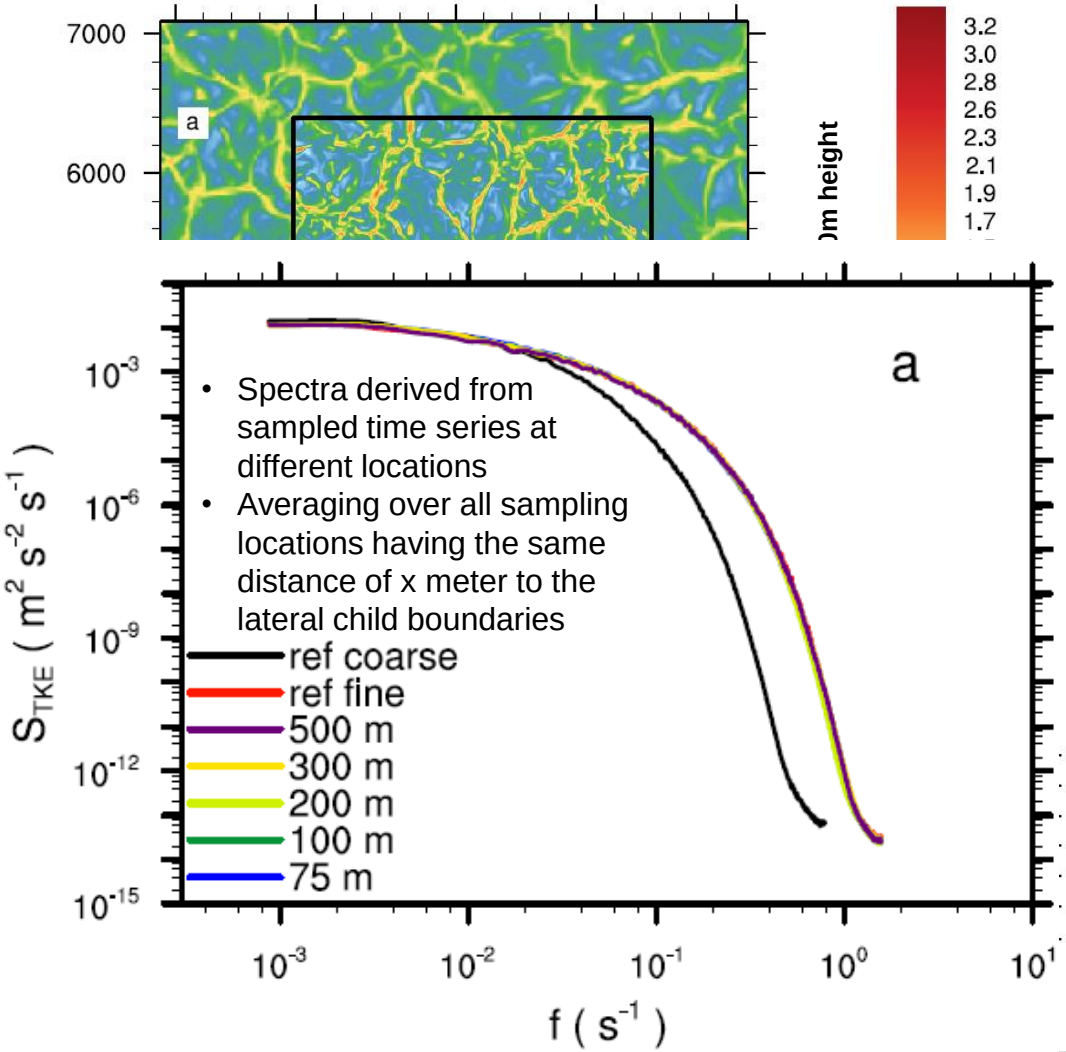
- Pure convective boundary layer with zero mean wind (homogeneously heated, flat terrain).
- Neutral boundary layer with background wind (purely shear-driven, flat-terrain).
- Neutrally-stratified urban boundary layer over a regular staggered arrangement of building cubes.

Pure convective boundary layer



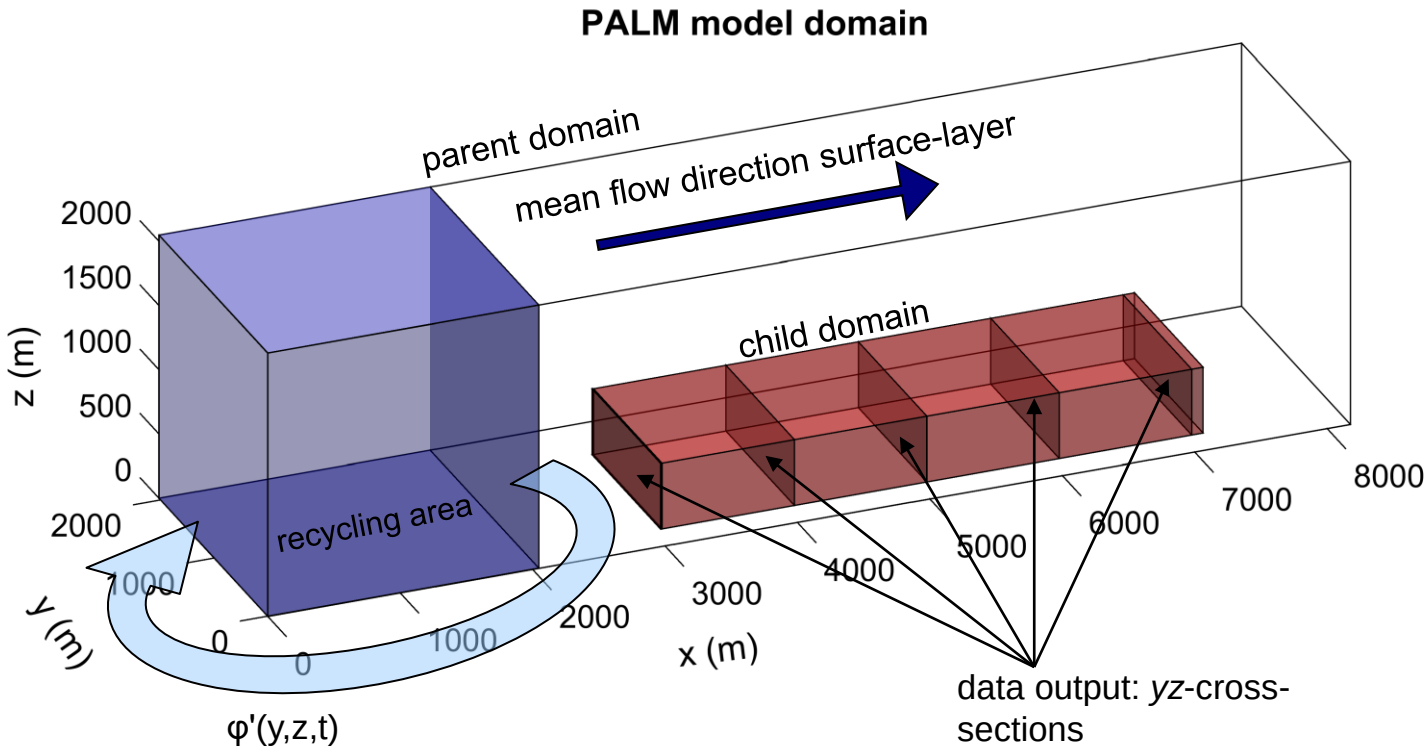
- Grid spacing ratio 20m:10m
- No discontinuities near boundaries in terms of shape and amplitude
- Finer structures within child domain with slightly stronger up/downdrafts
- Comparable size of hexagonal cells

Pure convective boundary layer



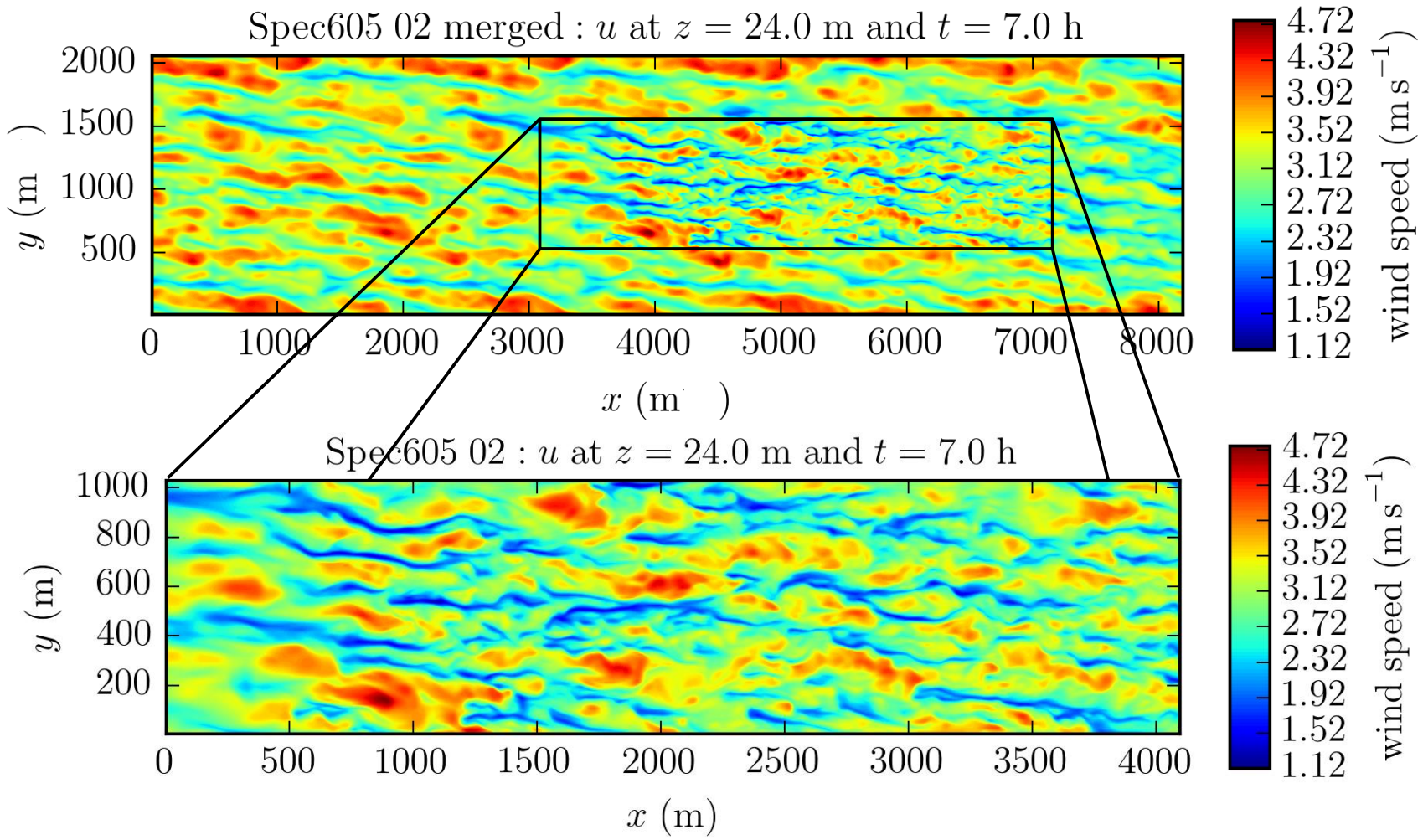
- Grid spacing ratio 20m:10m
- No discontinuities near boundaries in terms of shape and amplitude
- Finer structures within child domain with slightly stronger up/downdrafts
- Comparable size of hexagonal cells
- Better representation of spectral properties for fine-grid simulation
- Pure fine-grid simulation comparable to child solution independent of the distance to the boundaries
- In pure convective case (no background wind) almost no adjustment zone required since turbulence is mostly produced locally

Neutral boundary layer

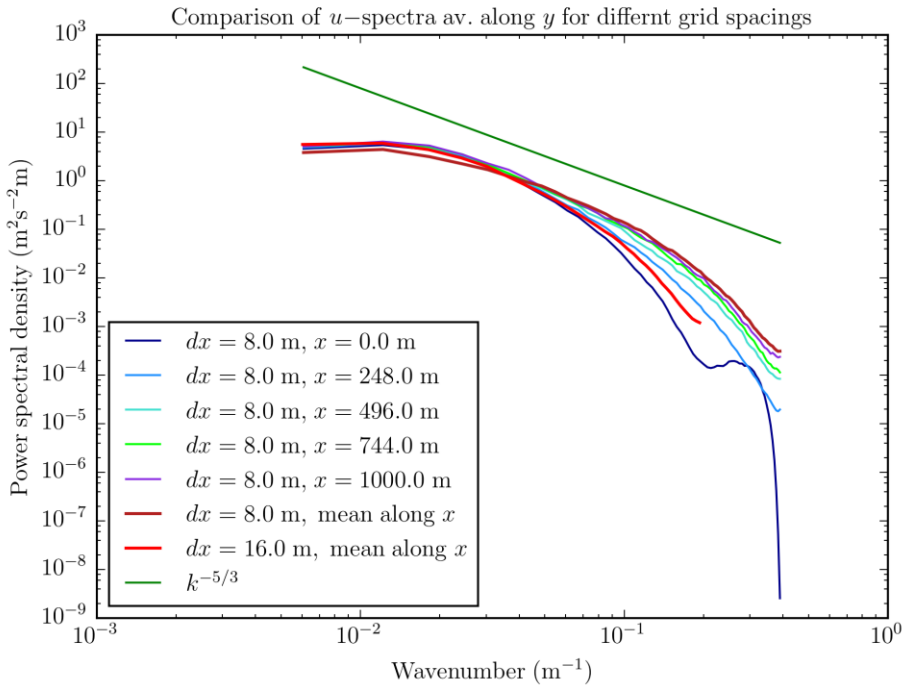


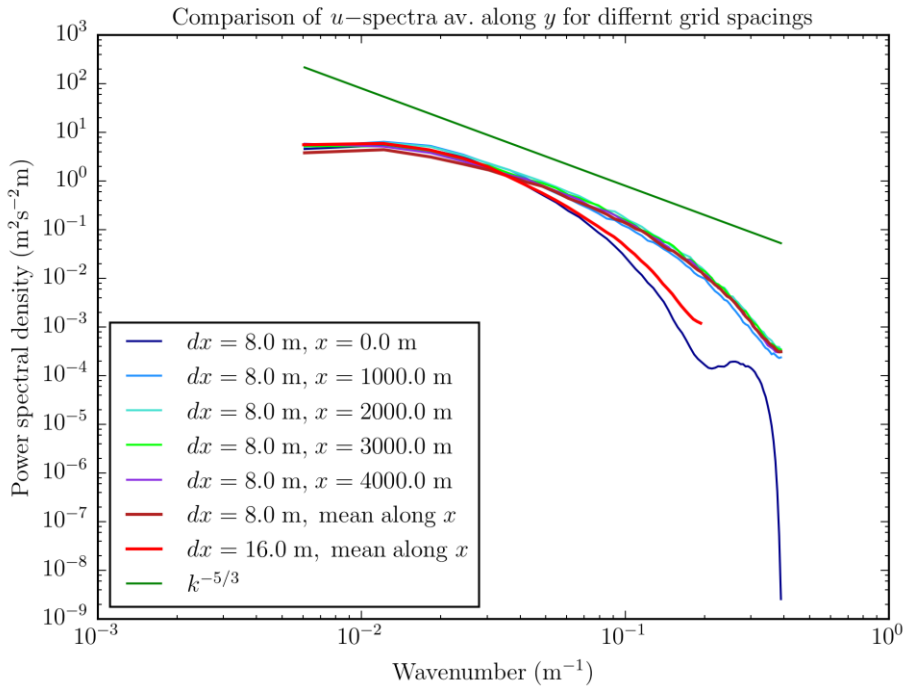
Grid for both domains: 512 x 128 x 64 gridpoints
Grid spacing: **16 m** (parent) and **8 m** (child)
Position nest: lower_left_x = 3072 m, lower_left_y = 512 m
PEs: each on 256
Coupling: Two-way

x-y cross-section of the streamwise velocity *u*

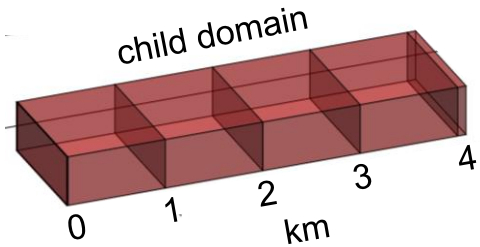


Spectra of *u*-component at different locations in streamwise direction





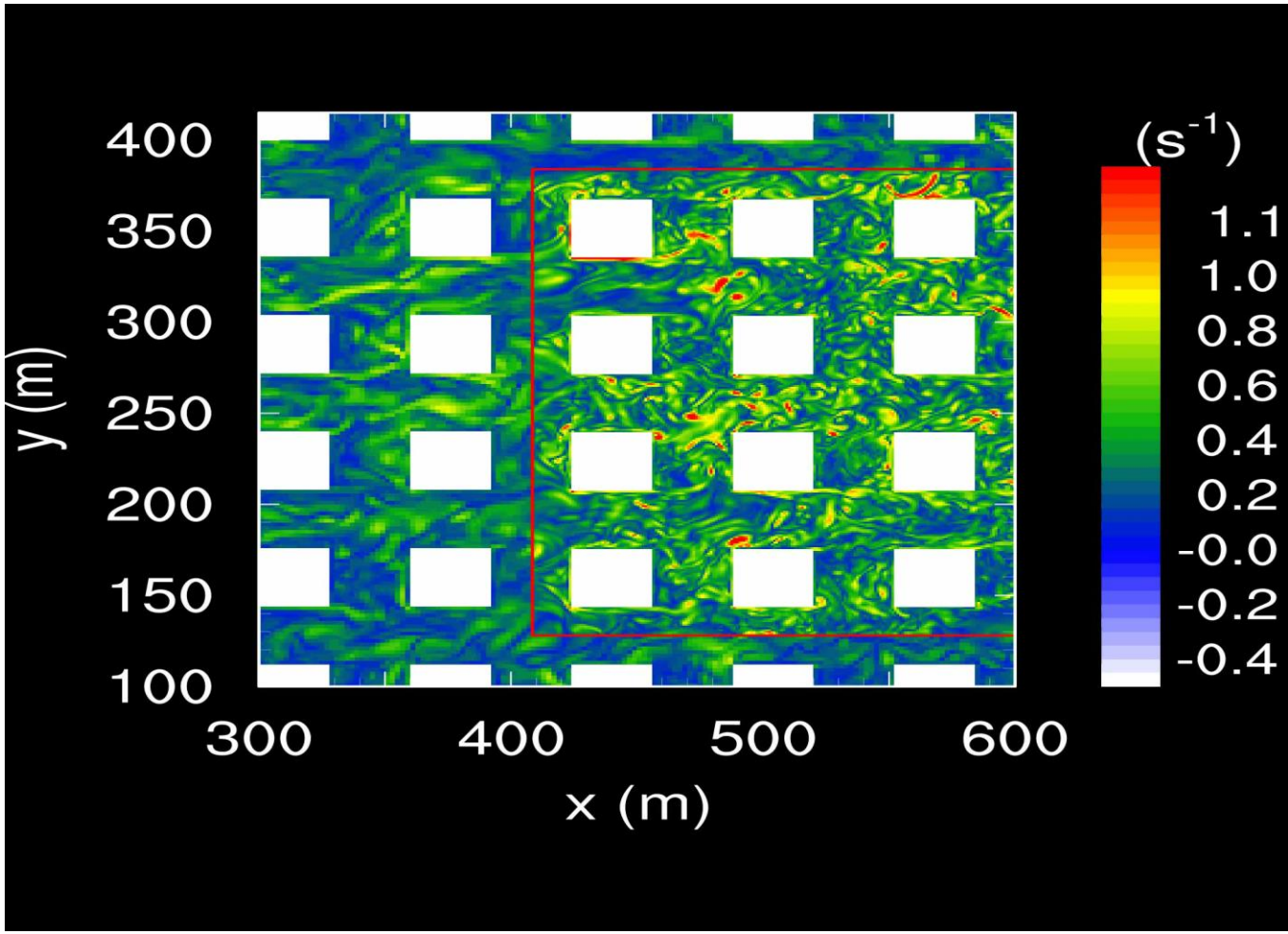
Spectra starts to overlap after more than 1 km behind the “inflow”
→ **Flow adjustment after 1-2 km**



Attention: Flow adjustment zone significantly increase with increasing grid spacing ratio

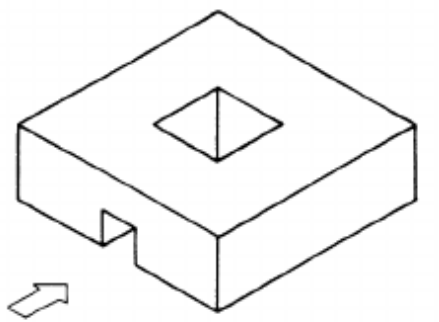
Urban boundary layer – pollutant dispersion on the city block scale

- Cutout of a nested pollutant dispersion simulation within an idealized city block
- Absolute value of curl of velocity is shown
- Background wind from left to right
- Child domain shows much more details of the flow
- Vortices are often generated at the building's edges
- Back flow behind buildings

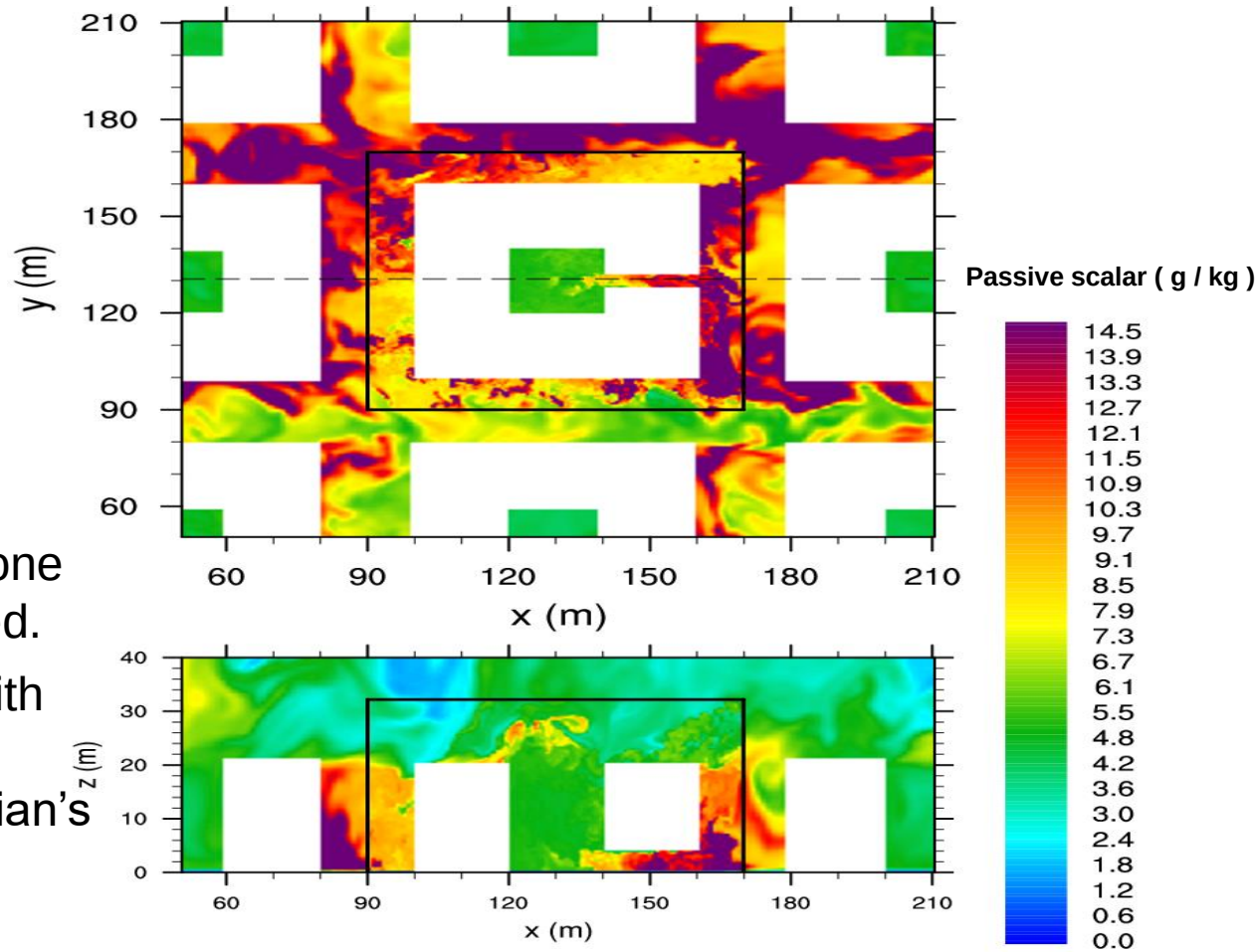


Urban boundary layer – pollutant dispersion on the building scale

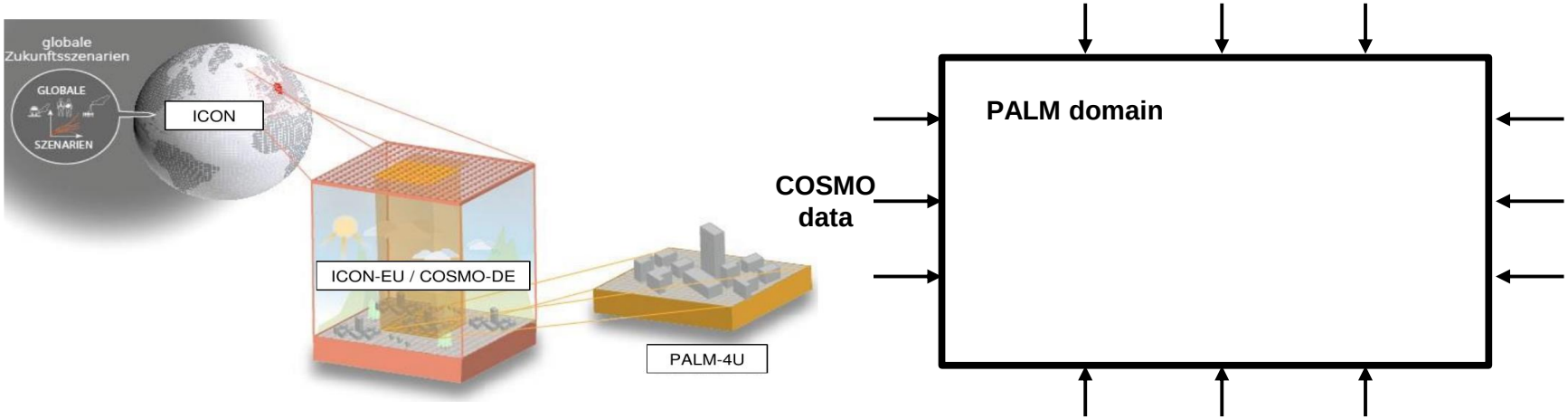
- How do courtyards and openings affect pollutant dispersion?



- The environment of just one single building was nested.
- Flow features together with concentration enable an evaluation of the pedestrian's well-being.

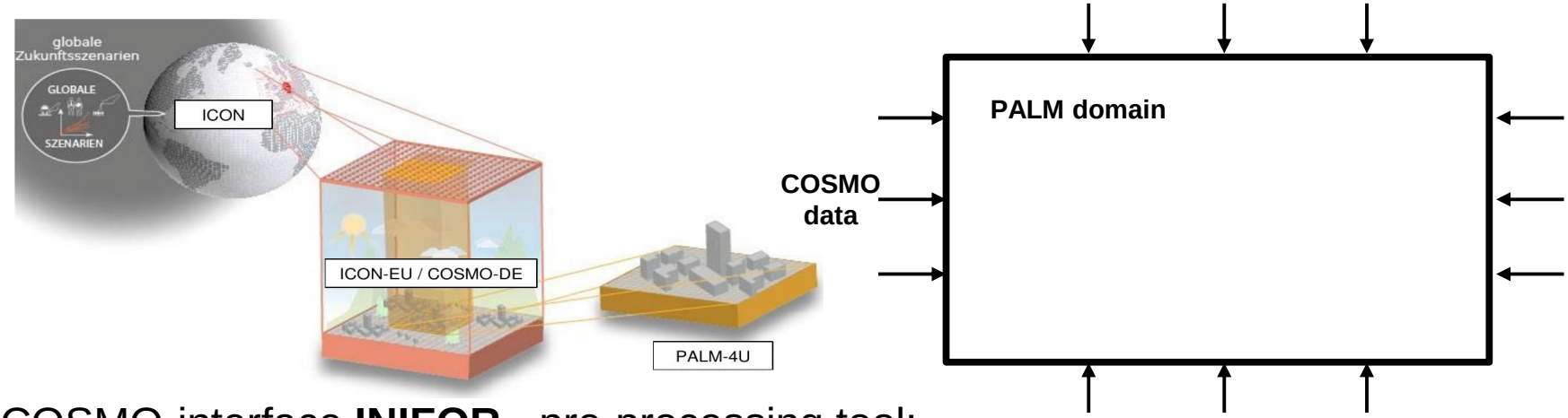


Mesoscale nesting within larger-scale models



- Idea: consider changes in synoptic conditions.
- Nest the model domain within larger-scale models, where the larger-scale model runs in advance.
- Provide pre-processed (offline) time-dependent data from, e.g. ICON, COSMO, or WRF model, at lateral and top boundaries of PALM domain via dynamic driver file.
- Boundary data is interpolated linearly in time.

Mesoscale nesting within COSMO model

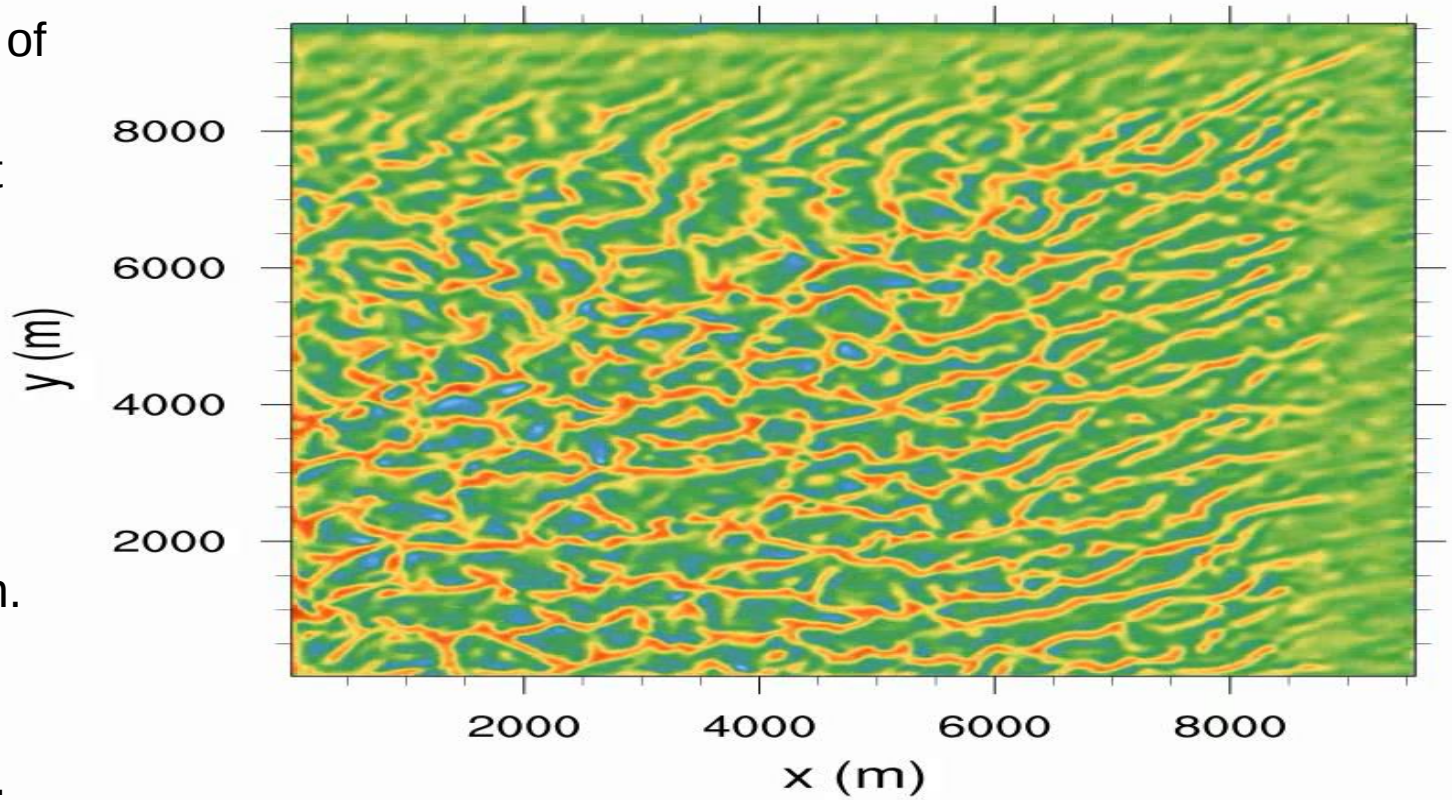


- COSMO-interface **INIFOR** - pre-processing tool:
  - **INI**tialization and **FOR**cing.
  - Interpolation of COSMO data onto Cartesian grid.
  - Provides initialization data of wind, temperature, humidity and soil temperature / moisture.
  - Provides time-dependent information on boundaries (lateral and top) for all relevant quantities.
  - Data is stored in “dynamic driver”, **`<ri>_dynamic`** .
  - Synthetic turbulence generator at lateral boundaries required to initiate turbulence

- **INIFOR** is now replaced by **promet**!

Mesoscale nesting in COSMO model

- COSMO forcing of 8 hrs
- Flow adjustment zone is clearly visible.
- Synoptic wind comes from northeast and turns to south eastern direction.
- Combination of band-like and cellular patterns.



Remarks

- **Open points self nesting:**
  - CBL test cases show secondary circulation in time-averaged fields with an upward motion in the child domain and downward motions at the child boundaries (is an inherent feature of the nesting).
  - Influence of coupling mode (one-way, two-way) should always be analyzed in detail.
  - **For most applications, one-way coupling is recommended.**
  - The RANS-RANS and RANS-LES nesting are not sufficiently tested and should be used with care.
  - Particle nesting needs further testing especially regarding the treatment of stochastic subgrid-scale particle speeds.

- For further information see:
  https://palm.muk.uni-hannover.de/trac/wiki/doc/tec/nesting
  https://doi.org/10.5194/gmd-14-3185-2021 (Self-nesting - technical paper)
  https://doi.org/10.5194/gmd-14-5435-2021 (Mesoscale nesting - technical paper)

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